

IFTESHA NAJNIN

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Sr. AI Engineer

Senior AI Engineer specializing in Generative AI, Agentic AI, Large Language Models (LLMs), and enterprise AI platforms. Experienced in designing and productizing scalable AI solutions, multi-agent systems, RAG applications, and AI accelerators across enterprise and open-source ecosystems. Passionate about translating cutting-edge AI research into production-ready solutions that solve complex business challenges and deliver measurable impact through intelligent automation and innovation.

TECHNICAL SKILLS

- Programming Languages: Python, R, SQL
- Databases: SQL-based management, NoSQL
- Analytical Skills: Statistics, Linear Algebra, Predictive Modeling, Data Visualization
- AI Architectures: Machine Learning, Deep Learning, NLP, Generative AI, LLMs, Time Series Analysis, Prompt Engineering
- Libraries & Frameworks:
 - Data Analysis & Visualization: Numpy, Pandas, Seaborn, Matplotlib, Plotly
 - Machine Learning & Generative AI: Tensorflow, PyTorch, Hugging Face, LangChain, Scanpy, Wandb
 - Web Development: Streamlit, FastAPI, Gradio, Chainlit
- Tools & Platforms: Git, CI/CD, Docker, Tableau, Azure
- Operating Systems: Linux, Windows, MacOS

WORK EXPERIENCE

Blue Yonder

Sr.AI Researcher/ Engineer

May 2025- Present

Architecting enterprise-grade Generative AI platforms and agentic systems to accelerate AI adoption across Blue Yonder. Leading the design, development, and productization of reusable AI agent frameworks, low-code agent development capabilities, and enterprise integrations in collaboration with Product Development teams.

AgentSmith – Enterprise Multi-Agent Platform

Tools Used: Python, FastAPI, Azure OpenAI, LangChain, LangGraph, BY SDK, Azure, CI/CD

Roles & Responsibilities:

- Co-invented AgentSmith, a low-code/no-code AI agent platform that reached the Crystal Ball Innovation Finals and was later integrated into Blue Yonder Agent Marketplace.
- Architected a modular multi-layer platform with reusable SDKs, accelerators, orchestration, and marketplace components.
- Led the development of Smart Builder, enabling natural language-driven agent creation, instruction generation, and intelligent tool recommendations.
- Designed backend APIs and orchestration services for agent lifecycle management and reusable AI workflows.
- Collaborated with Product Development teams to productize AgentSmith and align it with the enterprise roadmap.
- Presented architecture and product demos to executive leadership and AI CoE forums.
- Contributed to a reusable platform designed to support 500K+ AI agent executions per month while reducing custom development effort by ~50% through shared accelerators and services.

Agent Marketplace

Tools Used: Azure OpenAI, FastAPI, LangChain, BY SDK, GitHub, Docker, Azure Container Apps, CI/CD

Roles & Responsibilities:

- Integrated Smart Builder into Blue Yonder's Agent Marketplace to enable AI-driven agent generation.
- Built APIs for automated agent blueprint, instruction, and tool recommendation generation.
- Implemented streaming, reusable agent templates, and orchestration capabilities for enterprise deployment.

TAM Persona Agent – Microsoft & Blue Yonder Co-build | *UAT. Production Oct 2026*

Tools Used: Azure AI Foundry, Azure OpenAI, FastAPI, Multi-Agent Systems, RAG, LangChain, CI/CD • Azure

- Led the technical design and solution architecture for Blue Yonder's TAM Persona Agent as part of a strategic Microsoft co-build initiative.
- Designed and validated multi-agent orchestration, memory, context management, routing, and persona-driven agent architectures for enterprise AI workflows.
- Partnered with Microsoft architects and business stakeholders to translate TAM requirements into scalable, production-ready AI agent solutions.
- Designed and delivered multiple enterprise agents, including Upgrade Analyzer, Lead Analyzer, and Escalation Assist, leveraging RAG, semantic search, and multi-source enterprise data retrieval.
- Built Python-based agent services and integrated Azure AI Foundry, Microsoft Teams, and enterprise deployment pipelines.
- Established deployment architecture, authentication, security, CI/CD, and production readiness for enterprise-scale AI applications.
- Led technical workshops, architecture reviews, BRD refinement, and cross-functional collaboration across Microsoft, Product Development, AI CoE, and business teams.

Aganitha Cognitive Solutions

Full Stack Data Scientist -1

Feb 2023 - May 2025

LLM Agent for Sc-RNA Data Analysis Leveraging Chain of Thought Framework within LLMs.

Tools Used: OpenAI, LangChain, HuggingFace, Streamlit, Anndata, Scanpy, Docker, Chainlit, LangSmith, Docker

- Built a conversational ReAct agent that lets biologists run scRNA-seq analyses in natural language, with tool and database integrations for live data access.
- Extended it into a multi-agent framework and added a response-caching layer that materially reduced inference cost; instrumented with LangSmith for evaluation and debugging.

Antibody Titer Prediction Using Time Series Analysis

Tools Used: PyTorch, MlFlow, Sklearn, Numpy, Pandas, Matplotlib, Postgres, Streamlit, Docker

- Built an LSTM-based time-series forecasting model for antibody titer prediction, achieving 94% accuracy through feature engineering, hyperparameter optimization, and custom evaluation metrics.
- Developed an end-to-end prediction pipeline with missing-value interpolation, robust validation, and deployed the solution as an interactive Streamlit application.

GeneFormer: Advancing Biological Research with AI

Tools Used: HuggingFace, Transformer, Geneformer, Docker, Matplotlib, Seaborn, Pandas, Docker

- Fine-tuned Geneformer, a BERT-based foundation model for single-cell genomics, to improve biological data analysis across downstream prediction tasks.
- Optimized model performance through hyperparameter tuning and applied in-silico perturbation and classification techniques to derive gene-level insights.

Odia Generative AI

Sr.AI Researcher

Feb 2026 - Present

- Building an Odia AI ecosystem from datasets, OCR evaluation, and model benchmarking for a low-resource Indic language.
- Created and open-sourced one of the first Odia OCR benchmark datasets and fine-tuned Qwen2.5-VL with LoRA for high-accuracy Odia OCR.
- Developed Indic OCR Eval, introducing GER, CAGER, and Indic-aware OCR evaluation metrics for low-resource languages.
- Released datasets, models, and evaluation tools on Hugging Face to support reproducible research and multilingual AI.

HW Saver LLP

Subject Matter Lead (Math & Phy)

Oct 2019 - Apr 2022

Developed deep expertise in mathematical modeling, problem-solving, and analytical thinking. These skills now enhance my work as a Data Scientist, enabling me to design robust algorithms, interpret complex datasets, and deliver data-driven insights effectively.

EDUCATION

Master of Science (Physics) | 2017 - 2019

NIT, Srinagar, J&K.

CGPA - 9.2

Subjects: Advance Mathematics, Statistical Methods, Computational Physics, Electronics & Communication, Research Methodology

PG Diploma in Data Science | 2022-2023

AlmaBetter

ACTIVITIES

- Attended Dataverse Hack- Nov organized by Analytics Vidya.
- Attended ML Hackathon- August organized by Hacker Earth.
- Attended AstroSprint-Astronomy from Archival Data by MANUU, HYD

ACHIVEMENTS

- Received GEM Award- 2025 in BY
 - AIR-1388 in GATE-DA-2025
 - Achieved top 2% Approach in ML Hackathon Aug-2022.
 - Gold Medalist MSc. PHY (NIT, Srinagar)
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